

Kintan Saha

Third-year (Rising Junior) Undergraduate Student
B.Tech. (Mathematics and Computing)
Indian Institute of Science, Bengaluru, India

 sahakintan.github.io/myportfolio
 kintansaha@iisc.ac.in / saha5kintan@gmail.com
 91-9007471396
 scholar.google.com
 [linkedin.com](https://www.linkedin.com)

RESEARCH INTERESTS

My research interests lie in **Reinforcement Learning**, with a focus on **Stochastic Approximation** methods and establishing theoretical guarantees for learning algorithms, as well as in theoretical aspects of generative models, particularly **diffusion models**. I am also interested in **3D Computer Vision**, especially in 3D Scene Reconstruction and Novel View Synthesis.

EDUCATION AND RESEARCH TRAINING

Indian Institute of Science, Bengaluru Aug'23 – Present (Expected Graduation - Jul'27)
B.Tech. in Mathematics and Computing. GPA: 9.4/10.0 (till 5th semester). Within Top 5% of cohort

Key Courses Taken

[COMPLETE LIST](#)

Real Analysis ; Measure-Theoretic Probability ; Stochastic Calculus ; Stochastic Processes ; Concentration Inequalities ; Multi-Armed Bandits ; Stochastic Approximation ; Reinforcement Learning ; Machine Learning ; Optimization Theory ; Generative Models ; Nonlinear Systems and Control

Schools - Maa Bharti Sr Sec School, Kota / Delhi Public School Ruby Park, Kolkata Apr'09 – Jul'23
Senior Secondary (97.8%), Secondary (95.8%)

Achievements: [JEE Advanced](#) (AIR 338), [JEE Main](#) (AIR 87), [WBJEE](#) (AIR 7), Top 1% in [NSEP](#), [NSEC](#), [NSEA](#)

RESEARCH EXPERIENCE

[PROJECT DETAILS](#)

THEORETICAL MACHINE LEARNING

Finite-Time Guarantees for Multi-Timescale Stochastic Approximation and Hierarchical Reinforcement Learning | Nov 2025 – Present

- Developing finite-time analysis of multi-timescale stochastic approximation, with applications to hierarchical reinforcement learning, in collaboration with [Prof. Debabrota Basu \(INRIA SCOOL\)](#)

Stochastic Analysis on Graphons | Nov 2025 – Present

- Studying gradient-flow analysis and convergence properties on the space of graphons, in collaboration with [Prof. Anant Raj \(IISc\)](#).

REINFORCEMENT LEARNING

Reliable Policy Iteration [arXiv](#) | May 2025 – Sep 2025

- Collaborated with [Prof. Aditya Gopalan \(IISc\)](#) and [Prof. Gagan Thoppe \(IISc\)](#) to enhance the [Reliable Policy Iteration \(RPI\)](#) framework.
- Integrated RPI with state-of-the-art deep reinforcement learning algorithms (PPO, TD3, DDPG) and evaluated performance across diverse environments, including Atari, MuJoCo, and MiniGrid.
- Established new baselines and demonstrated state-of-the-art improvements using RPI-augmented algorithms; additionally showed robustness of RPI across multiple function approximation classes. Accepted for publication at the [Indian Control Conference 2025](#).

Towards Reliable, Uncertainty-aware Alignment [arXiv](#) | Jan 2025 – April 2025

- Collaborated with [Prof. Aditya Gopalan](#) to develop an uncertainty-aware policy optimization framework which provably reduces policy degradation.
- Extended the method to **LLM alignment** by modifying PPO to incorporate reward uncertainty in **RLHF** pipelines.

<i>Dataset Used</i>	<i>Tools / Libraries Applied</i>
Alpaca-Eval&mt-bench	HuggingFace Transformers, TRL and Datasets, Matplotlib, NumPy

COMPUTER VISION

Feed-Forward Deblurring in 3DGS | May 2025 – Jul 2025

- Collaborated with [Prof. Venkatesh Babu \(Vision and AI Lab, IISc\)](#) to design a **scene-agnostic deblurring module** for 3D Gaussian Splatting (3DGS) to be used in feed-forward models.

<i>Dataset Used</i>	<i>Tools / Libraries Applied</i>
ACID, Re10k	PyTorch, Lightning, Hydra

MACHINE LEARNING

- Unsupervised Domain Adaptation (UDA)** [\[Project Report\]](#) [\[Project Code\]](#) | Jan 2025 – April 2025
- **Implemented** and **benchmarked** multiple SOTA UDA algorithms: DANN, CORAL/DeepCORAL, MMD, DSN, and Asymmetric Tri-Training as part of a course project under [Prof Chiranjib Bhattacharya](#).
 - Evaluated cross-domain generalization on **diverse datasets**: MNIST, MNIST-M, SVHN, Office31, and Amazon Reviews.
 - Applied **adversarial training** with Gradient Reversal Layer (GRL), domain-invariant representation learning, and theoretical metrics like H-divergence.
 - Achieved **performance** within 5% of paper-reported results; analyzed key failure modes in domain classification due to activation function and network design choices.

<i>Dataset Used</i>	<i>Tools / Libraries Applied</i>	<i>Visualization Produced</i>
MNIST, MNIST-M, SVHN, Amazon Reviews	PyTorch, Matplotlib, NumPy	Domain classification plots, t-SNE

QUANTITATIVE RESEARCH

- Selected from the **top 1%** nationwide for [Optiver FutureFocus - 2025 program](#)
- Completed an intensive one-week program focused on quantitative trading workflows and applying data-driven modeling on in-house stock market datasets.
- Focused on preprocessing, visualization, and exploratory analysis for **alpha strategy design** using Pandas, NumPy and Matplotlib.

COMPUTER SCIENCE EDUCATION

HinglishEval: Evaluating Effectiveness of Code-generation Models on Hinglish Prompts [\[Paper\]](#) [\[Code\]](#) | Jan 2024 – June 2024

- Collaborated with [Prof. Viraj Kumar](#) to evaluate code-generation LLM performance on multilingual code-generation using a translated HumanEval dataset.
- Designed evaluation pipeline to analyse results under the **Item Response Theory (IRT)** framework.
- Accepted for publication at [ACM COMPUTE 2024](#); part of a broader effort to adapt LLMs for CS101 teaching in Indian classrooms.

<i>Dataset Used</i>	<i>Tools / Libraries Applied</i>
HumanEval	HuggingFace Transformers and Datasets, Matplotlib

PUBLICATIONS/ PREPRINTS

1. [Towards Reliable, Uncertainty-aware Alignment](#) (preprint)
2. [Reliable Policy Iteration: Performance Robustness Across Architecture and Environment Perturbations](#), ICC 2025
3. [HinglishEval: Evaluating the Effectiveness of Code-Generation Models on Hinglish Prompts](#), COMPUTE 2024

TECHNICAL SKILLS

Programming	Python, Shell scripting, Conda/Miniconda
ML / Deep Learning	PyTorch, HuggingFace Transformers, TRL, Diffusers, W&B, Hydra, OpenCV, Scikit-learn, XGBoost
Reinforcement Learning	Stable-Baselines3, OpenAI Gym (<i>Atari</i> , <i>MuJoCo</i> , <i>MiniGrid</i>)
Data Analysis & Viz.	Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit

SELECTED TALKS

[COMPLETE LIST](#)

1. [Presentation at Indian Control Conference 2025 of my paper](#)
2. [Presentation of course project for Topics in Stochastic Approximation Algorithms \(E1 396\)](#)
3. [Presentation of course project for Concentration Inequalities \(E2 207\)](#)
4. [Presentation of course project for Theory of Multi-armed Bandits \(E1 240\)](#)
5. [Presentation on Diffusion and Flow based models](#)
6. [Presentation of course project for UMC203 - AI and ML](#)
7. [Presentation of course project for E1 277 - Reinforcement Learning](#)
8. [Presentation at ACM COMPUTE 2024 of my paper](#)

VOLUNTEERING

Databased (IISc UG CS Club) — Senior Core Committee Member
Organizer, [ML Theory Series](#); curated seminar topics and mini-courses introducing theoretical machine learning. Served as an instructor for the [Reinforcement Learning](#) mini-course.

Rhythmica (IISc Music Club) — Co-Convener
Organized concerts, open mics, and classes. Experienced guitarist with 10+ years of [musical journey](#).